

Design: Mortar Transfer for Impedance-Overlap Schwarz on Non-Conforming Meshes

Alejandro Mota

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1 Problem statement

The impedance-overlap transmission condition

$$\mathbf{t}_k - \mathbf{t}_j + \mathbf{Z}(\dot{\mathbf{u}}_k - \dot{\mathbf{u}}_j) + \alpha(\mathbf{u}_k - \mathbf{u}_j) = \mathbf{0} \quad \text{on } \Gamma_k \quad (1)$$

is now exactly conservative on conforming meshes (0.08% PU-energy error over 1 ms with the consistent partner traction, matching the exact-DBC reference). On the non-conforming cantilever pair ($h = 8.47$ vs. 6.35 mm, 4:3) the net error also looks small ($\pm 1\%$), but the interface-work instrumentation shows that this is a cancellation of two error channels, each of which moves roughly *half the system energy per millisecond*:

1 ms, II, PU metric	conforming	non-conforming h	non-conforming $h/2$
dashpot (spurious) work	−42.6 J (2.8%)	−606.5 J (40%)	−598.5 J (40%)
traction-channel work	—	+677 J (45%)	+768 J (51%)
net PU-energy error	1.5% → 0.08%	−0.75% (growth)	+1.5% (loss)
mean velocity jump (RMS)	8.9 m/s	40 m/s (growing 15→60)	27 m/s

Three facts drive the design. (i) The channels do *not* shrink under mesh refinement (dashpot work unchanged at $h/2$), so resolution cannot fix the interface; the pointwise-interpolation transfer feeds the condition an $O(1)$ error on mesh-scale content. (ii) The net error is the difference of two large channels and even flips sign between h and $h/2$ — it is not robust, which is the production fragility. (iii) The velocity jumps grow through the run: each interface pass leaves mesh-scale residue that reverberates, a bounded version of the same feedback that makes DBC overlap coupling exponentially unstable.

The theoretical requirements are Gander–Halpern– Nataf’s three conditions for discrete energy convergence of waveform-relaxation coupling on nonmatching grids: (i) Robin (not Dirichlet) transmission, (ii) consistent boundary-cell discretization of the transmission operator, (iii) an L^2 -contractive intergrid transfer. The impedance-overlap BC satisfies (i); its weak (surface-mass-weighted) assembly satisfies (ii); pointwise interpolation violates (iii). Replacing the transfer with the L^2 (mortar) projection onto the receiving trace space satisfies all three simultaneously. Note that weak- L^2 enforcement of *Dirichlet* overlap coupling did not remove the DBC instability — consistent with the theory, since DBC violates (i) regardless of the transfer; the combination proposed here is the first that meets all three conditions.

2 Operator inventory (existing machinery)

All operators needed for the first stage already exist:

operator	code	status
$\mathbf{W}_k = \int_{\Gamma_k} \mathbf{N}_k \mathbf{N}_k^\top d\Gamma$	<code>get_square_projection_matrix</code>	already on the BC
$\mathbf{L}_{kj} = \int_{\Gamma_k} \mathbf{N}_k(\mathbf{x}) \mathbf{N}_j^{\text{vol}}(\mathbf{x})^\top d\Gamma$	<code>get_overlap_rectangular_projection_matrix</code>	exists (weak-DBC path); needs signature relaxed
$\mathbf{P}_{kj} = \mathbf{W}_k^{-1} \mathbf{L}_{kj}$	weak-DBC <code>dirichlet_projector</code>	same construction
surface-to-surface $\mathbf{R}_{kj}$	<code>get_rectangular_projection_matrix</code> (<code>project_point_to_side_set</code>)	exists (nonoverlap path); needed in stage 2
partner one-sided weak flux	<code>build_consistent_traction_patch!</code> , <code>consistent_partner_traction</code>	exists (conforming); generalized in stage 2

$\mathbf{L}_{kj}$ is assembled by quadrature over Ω_k 's boundary facets; at each quadrature point the partner *volume* element is located (`find_point_in_mesh`) and its shape functions evaluated — exactly the mortar cross-mass for an overlap boundary, which is interior to the partner and therefore has no partner surface mesh.

Two structural properties come for free. First, $\mathbf{P}_{kj} = \mathbf{W}_k^{-1} \mathbf{L}_{kj}$ is the $L^2(\Gamma_k)$ -orthogonal (Galerkin) projection onto Ω_k 's trace space, hence non-expansive in L^2 : GHN condition (iii) holds by construction, for any quadrature, because it is a projection. Second, partition of unity of the partner shape functions gives $\mathbf{L}_{kj} \mathbf{1} = \mathbf{W}_k \mathbf{1}$ for *any* quadrature rule, so constants (rigid translations) transfer exactly regardless of quadrature accuracy.

3 Design

3.1 Stage 1: mortar projection of the transferred fields

Replace pointwise interpolation of the three partner fields with the mortar projection, leaving everything downstream of the nodal values unchanged:

$$\hat{\mathbf{u}}_j = \mathbf{P}_{kj} \dot{\mathbf{u}}_j, \quad \hat{\mathbf{u}}_j = \mathbf{P}_{kj} \mathbf{u}_j, \quad \hat{\boldsymbol{\sigma}}_j = \mathbf{P}_{kj} \boldsymbol{\sigma}_j^{\text{rec}} \quad (\text{six Voigt components}), \quad (2)$$

then form the traction $\hat{\mathbf{t}}_j = \hat{\boldsymbol{\sigma}}_j \mathbf{n}_k$ and the tensor-impedance term nodally exactly as the current code does. This is a drop-in change: `apply_bc_detail`'s per-node interpolation loop becomes three (or nine, per component) small dense matrix–vector products with a projector precomputed at setup. The P/S-split tensor $\mathbf{Z}(\mathbf{n})$ continues to act at the receiving nodes, so no re-derivation of the self terms or the tangent is needed; the variational crime of nodal application is unchanged from the current scheme and identical on both branches.

Implementation touch points:

- **YAML:** new BC option `transfer:` with values `pointwise` (current behavior, initial default) and `mortar`.
- **Types:** field `mortar_projector::Matrix{Float64}` ($n_k \times N_j$, dense; boundary node count times partner node count, small for realistic interfaces) on `SolidMechanicsImpedanceOverlapSchwarzBound`

- **Setup** (`compute_impedance_overlap_schwarz_projectors!`): relax `get_overlap_rectangular_projection` signature to accept the impedance BC (it only touches fields both BC types share), build $\mathbf{L}_{kj}$, factorize the existing $\mathbf{W}_k$ (Cholesky) once, store $\mathbf{P}_{kj}$. Rebuilt on BC swaps, like the consistent-traction patch.
- **Evaluation**: factor the partner-field sampling shared by `apply_bc_detail` and `report_impedance_interface` into one helper that dispatches on the transfer mode, so the instrumentation always measures what the BC actually applies.
- **Precedence**: on node-aligned interfaces $\mathbf{L}_{kj}$ has a single unit entry per row and $\mathbf{P}_{kj}$ reduces to the identity; the consistent-traction patch (`partner traction: auto`) continues to take priority for the traction, and the kinematic projection becomes a no-op. The conforming limit is therefore automatic.

Quadrature note: the integrand of $\mathbf{L}_{kj}$ is only piecewise smooth on Γ_k (partner elements change within a receiving facet), so the default facet rule carries a consistency error. Because contractivity and constant-transfer hold for any rule, this error degrades accuracy, not stability. The cheap knob is an elevated or subdivided facet rule for the $\mathbf{L}_{kj}$ assembly only (setup cost); a true common-refinement (2D polygon overlay on the flat interface) is the exact alternative and is deferred until measurements justify it.

What stage 1 answers: how much of the 600 J churn is the *transfer* (interpolation aliasing of mesh-scale content) versus the *recovered-stress traction*. The conforming ladder bounds what recovery-quality traction can achieve ($\approx 1.5\%$ -class); if mortar transfer restores $O(h)$ scaling of the channels, refinement plus stage 2 closes the rest.

3.2 Stage 2: consistent traction across the interface

The conforming analysis showed the traction channel is the dominant residual once transfer is clean, and that nodal recovery cannot represent the traction carried by mesh-scale content. The consistent (one-sided partial assembly) traction requires a partner facet surface bounding a one-sided element patch; on a non-conforming interface Γ_k cuts through partner elements, so the patch does not exist on Γ_k itself. Two generalizations, in order of increasing cost:

2a: offset facet surface + surface mortar. Take $\tilde{\Gamma}_j$ = the partner facet surface immediately exterior to Γ_k (the boundary between the first exterior element layer and the rest; the patch-construction logic already classifies these elements). The consistent flux is exact on $\tilde{\Gamma}_j$ by the same partial assembly as the conforming case, tested with partner surface functions. Transfer the resulting weak traction to Γ_k 's trace space with the surface-to-surface mortar (`get_rectangular_projection_matrix` generalized to accept the internal surface in place of a side set, or an equivalent assembled operator), i.e. $\mathbf{W}_k \hat{\mathbf{t}}_j = \mathbf{R}_{k\tilde{j}} \mathbf{W}_{\tilde{j}}^{-1} \lambda_{\tilde{j}}$. The surface offset introduces an $O(h_j)$ modeling error ($\tilde{\Gamma}_j$ is within one partner element of Γ_k), but it acts on the *exact discrete flux* rather than on recovered stress; whether that trade wins is an experiment, not a theorem — measure before committing further.

2b: cut-cell partial assembly (flat interfaces). For a planar Γ_k (the common production configuration), integrate the partner's momentum residual over the plane-clipped parts of the cut elements, testing with Ω_k 's surface functions extended constant along the interface normal. This reproduces the conforming construction exactly in the limit and removes the offset error of 2a, at the cost of new machinery (hexahedron-half-space clipping and quadrature on the clipped polyhedra). Deferred unless 2a measures short of the target.

3.3 Validation and targets

The metric is the instrumentation’s channel decomposition, not the net energy: success is driving the churn itself down, since the current scheme already fakes a small net by cancellation.

1. **Conforming regression:** mortar transfer on the node-aligned pair must reproduce the pointwise results exactly ($\mathbf{P} = \mathbf{I}$); the consistent-traction 0.08% result must be unaffected. Extend test 103/104 asserts with the projector identity check.
2. **Non-conforming, stage 1:** dashpot work $\ll 600$ J at h , and restored $O(h)$ decay at $h/2$; velocity-jump growth curve flattened. Expected end state $\approx$ conforming recovered-stress class (few percent), now robust rather than cancellation-balanced.
3. **Non-conforming, stage 2:** traction channel reduced toward the consistent-traction class; net PU error $\lesssim 1\%$ with both channels individually small; sign and magnitude stable under refinement.
4. **Stability margin:** rerun the impedance-scale sweep — with contractive transfer the growth branch above scale 1 should soften, and the DBC-limit blow-up should set in later; this is the direct measure of production robustness.
5. **Cost:** setup $O(n_{\text{facets}} \times n_{\text{qp}})$ point locations (same as the weak-DBC path); per-iteration cost *decreases* (dense mat-vec replaces per-node interpolation loops; no recovery solve once stage 2 lands).

4 Summary

Stage 1 is a low-risk, high-information change that reuses the square ($\mathbf{W}_k$) and rectangular ($\mathbf{L}_{kj}$) projection machinery verbatim: mortar-project $\dot{\mathbf{u}}_j$, $\mathbf{u}_j$, and $\boldsymbol{\sigma}_j^{\text{rec}}$ onto the receiving trace space and keep the rest of the impedance pipeline unchanged. It makes the transfer L^2 -contractive by construction, completing GHN’s three conditions for the first time in this code, and its measurements decide between the two stage-2 traction designs. The conforming limit and the consistent-traction path are preserved automatically.